

File

SYDNEY GIRLS HIGH SCHOOL



MATHEMATICS EXTENSION 1

Assessment Task 2 for 2008 HSC

March 6th, 2008

Reading Time 5 minutes
Time allowed: 75 minutes

Topics: Trigonometry

DIRECTIONS TO CANDIDATES:

- Attempt all questions
- Questions are of equal value
- There are 4 questions with part marks shown
- All necessary working must be shown. Marks may be deducted for careless or badly arranged work
- Board approved calculators may be used
- Answers may be left in either exact form or correct to three significant figures
- Write on one side of the paper only
- Note the Integrals: $\int \sin(ax + b) dx = -\frac{1}{a} \cos(ax + b) + c$

$$\int \cos(ax + b) dx = \frac{1}{a} \sin(ax + b) + c$$

Question 1 (15 marks)

Marks

a) Evaluate :

i. $3 \lim_{x \rightarrow 0} \frac{\tan x}{x}$

1

ii. $\lim_{x \rightarrow 0} \frac{\sin 5x}{4x}$

2

b) Find, correct to the nearest degree, the acute angle between the lines
 $x + y - 4 = 0$ and $y = 2x + 1$

4

c) Differentiate with respect to θ

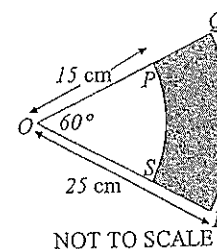
i. $y = 4 \cos^3 \theta - 3 \cos \theta$

2

ii. $y = \sec \theta - \cot \theta$

2

d)



PS and QR are arcs of concentric circles with O as centre.
Calculate in terms of π , the perimeter of the shaded region PQRS.

4

Question 2 (15 marks) (Start a new page)

- | | |
|---|-------|
| a) Find the exact value of $\tan \frac{5\pi}{6}$, expressing your answer in surd form with a rational denominator. | Marks |
| | 2 |
| b) | |
| i. Write out the expansion of $\sin(x+y)$ | 1 |
| ii. Hence show that $\sin(x + \frac{\pi}{2}) = \cos x$ | 1 |
| c) Find: | |
| i. $\int_0^{\pi} \cos 3x dx$ | 2 |
| ii. $\int_0^{\pi/2} \sin^2 2x dx$ | 2 |
| d) | |
| i. Express $\sqrt{3} \sin x + \cos x$ in the form $C \sin(x + \alpha)$ where $C > 0$ and $0 \leq \alpha \leq \frac{\pi}{2}$ | 2 |
| ii. Hence or otherwise, solve the equation $\sqrt{3} \sin x + \cos x = 1$ for $0 \leq x \leq 2\pi$ | 2 |
| e) Prove the identity $\frac{\cos 2\theta}{\sin \theta + \cos \theta} = \cos \theta - \sin \theta$ | 3 |

Question 3 (15 marks) (Start a new page)

Marks

- | | | |
|----|---|---|
| a) | (i) Express $\tan(\alpha - \beta)$ in terms of $\tan \alpha$ and $\tan \beta$. | 1 |
| | (ii) Use the result from (i) to find the exact value of $\tan 15^\circ$ | 3 |
| b) | Let $t = \tan \frac{\theta}{2}$ and starting with $\tan \theta = \tan(\frac{\theta}{2} + \frac{\theta}{2})$, show that $\tan \theta = \frac{2t}{1-t^2}$. | 3 |
| c) | Use the 't' results to solve $3\cos \theta + 4\sin \theta = -3$ for $0^\circ \leq \theta \leq 360^\circ$ giving your answer correct to the nearest minute. | 4 |
| d) | Find the exact value of the volume of solid of revolution formed when the region bounded by curve $y = \sin x$, the x-axis and the line $x = \frac{\pi}{4}$ is rotated about the x-axis. | 4 |

Question 4 (15 marks) (Start a new page)

Marks

- a) Find the equation of the tangent to the curve $y = \frac{1}{2} \cos 4x$ at the point
where $x = \frac{\pi}{8}$. 3
- b) A sector of a circle has an area of 24 cm^2 . Find the radius if the angle at the
centre of the circle is 3 radians. 2
- c) Solve the equation $\tan^2 2x = 1$ for $0 \leq x \leq \pi$. 3
- d) Solve the equation $2 \cos^2 x - \sin x - 1 = 0$ for $0 \leq x \leq 2\pi$. 3
- e) (i) Draw a neat sketch of $y = \frac{1}{2} \sin 2x + 2$ for $0 \leq x \leq \pi$ showing all
important features. 2
- (ii) On the same graph sketch $y = 1$. 1
- (iii) How many solution of $\frac{1}{2} \sin 2x + 1 = 0$ are there for $0 \leq x \leq \pi$? 1

S.C.H.S. Ext ① - Task ⑤ March 2008

Question 1 (15 marks)

A ✓
B ✓
C ✓
D ✓
E ✓
F ✓
G ✓
H ✓

a) i) $3 \lim_{x \rightarrow 0} \frac{\tan x}{x} = 3 \times 1$

$= 3$ ①

(ii) $\lim_{x \rightarrow 0} \frac{\sin 5x}{4x} = \frac{5}{4} \lim_{x \rightarrow 0} \frac{\sin 5x}{5x}$

$= \frac{5}{4}$ ②

b) $x + y - 4 = 0$ $y = 2x + 1$

$y = -x + 4$ $m_2 = 2$

$m_1 = -1$

$\tan \theta = \left| \frac{m_1 - m_2}{1 + m_1 m_2} \right|$

$= \left| \frac{-1 - 2}{1 + (-1 \times 2)} \right|$

$= \left| \frac{-3}{1 - 2} \right|$

$\tan \theta = 3$

$\therefore \theta = 71^\circ 34'$

$\approx 72^\circ$ ④

c) i) $y = 4 \cos^3 \theta - 3 \cos \theta$

$\frac{dy}{d\theta} = 4 \cdot 3 \cos^2 \theta \cdot -\sin \theta + 3 \sin \theta$

$= 3 \sin \theta - 12 \sin \theta \cos^2 \theta$ ②

(ii) $y = \sec \theta - \cot \theta$

$= \frac{1}{\cos \theta} - \frac{\cos \theta}{\sin \theta}$

c) (ii) $y = (\cos \theta)^{-1} - (\tan \theta)^{-1}$

$\frac{dy}{d\theta} = -1(\cos \theta)^{-2} \cdot -\sin \theta + 1(\tan \theta)^{-2} \times \sec^2 \theta$

$= \frac{\sin \theta}{\cos^2 \theta} + \frac{\sec^2 \theta}{\tan^2 \theta}$

$= \frac{\sin \theta}{\cos^2 \theta} + \frac{\cos^2 \theta}{\sin^2 \theta \cdot \cos^2 \theta}$

$= \frac{\sin \theta}{\cos^2 \theta} + \frac{1}{\sin^2 \theta}$ ②

$= \tan \theta \cdot \sec \theta + \operatorname{cosec}^2 \theta$

d) $\theta = 60^\circ$

$= \frac{\pi}{3}$

length of arc PS = $r\theta$

$= 15 \times \frac{\pi}{3}$

$= 5\pi \text{ cm}$

length of arc QR = $r\theta$

$= 25 \times \frac{\pi}{3}$

$= \frac{25\pi}{3}$

$PQ = SR = 25 - 15$

$= 10 \text{ cm}$

$\therefore \text{Perimeter of Shaded Region} = 5\pi + \frac{25\pi}{3} +$

$= \frac{40\pi}{3} + 20 \text{ cm}$

④

$$Q_2 \text{ a) } \tan\left(\frac{5\pi}{6}\right) = \tan\left(\pi - \frac{\pi}{6}\right)$$

$$= -\tan\frac{\pi}{6}$$

$$= \frac{-1}{\sqrt{3}} \text{ (1 1/2 marks)}$$

$$= -\frac{\sqrt{3}}{3} \text{ (2 marks)}$$

$$\text{b) i) } \sin(x+y) = \sin x \cos y + \cos x \sin y \text{ (1 mark)}$$

$$\text{ii) } \sin\left(x + \frac{\pi}{2}\right) = \sin x \cos \frac{\pi}{2} + \cos x \sin \frac{\pi}{2}$$

$$= \sin x \times 0 + \cos x \times 1$$

$$= \cos x$$

(1 mark)

$$\text{c) i) } \int_0^{\frac{\pi}{4}} \cos 3x \, dx = \left[\frac{1}{3} \sin 3x \right]_0^{\frac{\pi}{4}}$$

$$= \frac{1}{3} \sin \frac{3\pi}{4} - 0$$

$$= \frac{1}{3} \times \frac{1}{\sqrt{2}}$$

$$= \frac{1}{3\sqrt{2}} \text{ (1 1/2 marks)}$$

$$= \frac{\sqrt{2}}{6} \text{ (2 marks)}$$

$$\text{ii) } \int_0^{\frac{\pi}{2}} \sin^2 2x \, dx = \frac{1}{2} \int_0^{\frac{\pi}{2}} (1 - \cos 4x) \, dx$$

$$= \frac{1}{2} \left[x - \frac{1}{4} \sin 4x \right]_0^{\frac{\pi}{2}} \text{ (1 mark)}$$

$$= \frac{1}{2} \left[\frac{\pi}{2} - 0 \right] - [0]$$

$$= \frac{\pi}{4} \text{ (2 marks)}$$

$$\text{d) I) } C \sin(x+\alpha) = C (\sin x \cos \alpha + \cos x \sin \alpha)$$

$$\sqrt{3} \sin x + \cos x = C \cos x \sin x + C \sin x \cos x$$

$$\therefore C \cos \alpha = \sqrt{3} \text{ --- (1)}$$

$$C \sin \alpha = 1 \text{ --- (2)}$$

$$C^2 (\sin^2 \alpha + \cos^2 \alpha) = 3 + 1$$

$$C^2 = 4$$

$$C = 2$$

$$C > 0$$

$$\tan \alpha = \frac{1}{\sqrt{3}}, \alpha = \frac{\pi}{6}$$

$$\sqrt{3} \sin x + \cos x = 2 \sin\left(x + \frac{\pi}{6}\right) \text{ (2 marks)}$$

$$\text{II) } 2 \sin\left(x + \frac{\pi}{6}\right) = 1$$

$$\sin\left(x + \frac{\pi}{6}\right) = \frac{1}{2} \text{ (1 mark)}$$

$$x + \frac{\pi}{6} = \frac{\pi}{6}, \pi - \frac{\pi}{6}, \frac{13\pi}{6}$$

$$x = 0, \frac{2\pi}{3}, 2\pi \text{ (2 marks)}$$

e)

$$\frac{\cos 2\theta}{\sin \theta + \cos \theta}$$

$$= \frac{\cos^2 \theta - \sin^2 \theta}{\sin \theta + \cos \theta} \text{ (1 mark)}$$

$$= \frac{(\cos \theta + \sin \theta)(\cos \theta - \sin \theta)}{(\sin \theta + \cos \theta)} \text{ (2 marks)}$$

$$= \cos \theta - \sin \theta \text{ (3 marks)}$$

Q3 a) i) $\tan(\alpha - \beta) = \frac{\tan \alpha - \tan \beta}{1 + \tan \alpha \tan \beta}$

ii) $\tan 15^\circ = \tan(45^\circ - 30^\circ)$
 $= \frac{\tan 45^\circ - \tan 30^\circ}{1 + \tan 45^\circ \tan 30^\circ}$

Or:
 $\tan(60^\circ - 45^\circ) \text{ etc.}$
 $= \frac{1 - \frac{1}{\sqrt{3}} \times \frac{\sqrt{3}}{\sqrt{3}}}{1 + 1 \times \frac{1}{\sqrt{3}}}$
 $= \frac{(\sqrt{3}-1) \times (\sqrt{3}-1)}{\sqrt{3}+1 (\sqrt{3}-1)}$
 $= \frac{4 - 2\sqrt{3}}{2}$
 $= 2 - \sqrt{3}$

b) $\tan \theta = \tan\left(\frac{\theta}{2} + \frac{\theta}{2}\right)$
 $= \frac{\tan \frac{\theta}{2} + \tan \frac{\theta}{2}}{1 - \tan \frac{\theta}{2} \cdot \tan \frac{\theta}{2}}$
 $= \frac{2 \tan \frac{\theta}{2}}{1 - \tan^2 \frac{\theta}{2}}$
 $= \frac{2t}{1-t^2}$

c) $3 \cos \theta + 4 \sin \theta = -3$ for $0^\circ \leq \theta \leq 360^\circ$

$\sin \theta = \frac{2t}{1+t^2}$, $\cos \theta = \frac{1-t^2}{1+t^2}$

$\therefore \frac{3(1-t^2)}{1+t^2} + \frac{4(2t)}{1+t^2} = -3$

$3(1-t^2) + 8t = -3(1+t^2)$

$3 - 3t^2 + 8t = -3 - 3t^2$

$8t = -6$

$t = -\frac{3}{4}$

$\therefore \tan \frac{\theta}{2} = -\frac{3}{4}$ for $0^\circ \leq \frac{\theta}{2} \leq 180^\circ$

$\therefore \frac{\theta}{2}$ is an obtuse angle

$\frac{\theta}{2} = \tan^{-1}\left(-\frac{3}{4}\right)$

$= -36^\circ 52'$

but test $\theta = 180^\circ$ as
 't' method does not allow
 for such a solution.

When $\theta = 180^\circ$

LHS $= 3 \cos 180^\circ + 4 \sin 180^\circ$

$= 3 \times -1 + 0$

$= -3 \therefore \text{(True)}$

$= \text{RHS}$

$\therefore$ solutions are $180^\circ, 286^\circ 16'$

d) $V = \pi \int_a^b y^2 dx$

where $y = \sin x$

$\therefore y^2 = \sin^2 x$

$\therefore V = \pi \int_0^{\frac{\pi}{4}} \sin^2 x dx$

but $\sin^2 x = \frac{1}{2}(\cos 2x + 1)$

$\therefore V = \pi \times \frac{1}{2} \int_0^{\frac{\pi}{4}} (\cos 2x + 1) dx$

$= \frac{\pi}{2} \left[\frac{1}{2} \sin 2x + x \right]_0^{\frac{\pi}{4}}$

$= \frac{\pi}{2} \times \left(\frac{1}{2} \sin \frac{\pi}{2} + \frac{\pi}{4} \right)$

$= \frac{\pi}{2} \left(\frac{1}{2} + \frac{\pi}{4} \right)$

$= \frac{\pi}{4} + \frac{\pi^2}{8} \quad (u^3)$

$= \frac{\pi^2}{8} - \frac{\pi}{4}$

Question 4

a) $y = \frac{1}{2} \cos 4x$

$\frac{dy}{dx} = -2 \sin 4x$ ✓

at $x = \frac{\pi}{8}$, $m = -2 \sin \frac{\pi}{2} = -2$ ✓

$y = \frac{1}{2} \cos \frac{\pi}{2} = 0$ ✓

Eqn of tangent

$y - 0 = -2(x - \frac{\pi}{8})$

$y = -2x + \frac{\pi}{4}$ #

[3]

b) $A = \frac{1}{2} r^2 \theta$ ✓, $A = 24$, $\theta = 3$
 $24 = \frac{3}{2} r^2$

$r^2 = 16$

$r = 4$ cm ✓

[2]

c) $\tan^2 2x = 1$ $0 \leq x \leq \pi$

$\tan 2x = \pm 1$ ✓ $0 \leq 2x \leq 2\pi$

$2x = \frac{\pi}{4}, (\pi + \frac{\pi}{4}), (\pi - \frac{\pi}{4}), (2\pi - \frac{\pi}{4})$

$2x = \frac{\pi}{4}, \frac{5\pi}{4}, \frac{3\pi}{4}, \frac{7\pi}{4}$ ✓

$x = \frac{\pi}{8}, \frac{3\pi}{8}, \frac{5\pi}{8}, \frac{7\pi}{8}$ ✓ #

[3]

d) $2 \cos^2 x - \sin x - 1 = 0$ $0 \leq x \leq 2\pi$

$2(1 - \sin^2 x) - \sin x - 1 = 0$ ✓

$2 - 2 \sin^2 x - \sin x - 1 = 0$

$-2 \sin^2 x - \sin x + 1 = 0$

$2 \sin^2 x + \sin x - 1 = 0$

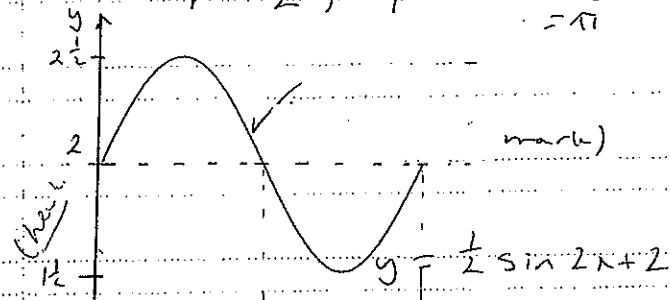
$(2 \sin x - 1)(\sin x + 1) = 0$

$\sin x = \frac{1}{2}$ or $\sin x = -1$ ✓

$x = \frac{\pi}{6}, \frac{5\pi}{6}, \frac{3\pi}{2}$ ✓ #

[3]

e) i) $y = \frac{1}{2} \sin 2x + 2$
 amp = $\frac{1}{2}$, period = $\frac{2\pi}{2} = \pi$, Vertical Shift = 2 ✓



[2]

ii) $y = 1$ ✓

[1]

i) Award 1st mark for correct amp, period, V shift
 Award 1 mark for any two of above correct
 Award second mark for correct sketch with correct amp, period, V shift. OR 1 mark for incorrect sketch (wrong period, amp etc of equivalent difficulty)

ii) as above

iii) $\frac{1}{2} \sin 2x + 1 = 0$

ie $\frac{1}{2} \sin 2x + 2 = 1$
 this is solved by finding the intersection of $y = \frac{1}{2} \sin 2x + 2$ and $y = 1$

As the two graphs do not intersect there are no solutions
 [No carry through here as it changes the intent of the question.] ✓

[1]

Total [15]